

FORTALEZA, Brazil

WMO: 823980

Lat: 3.776S

Lon: 38.533W

Elev: 82

StdP: 14.65

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	73.0	73.4	61.0	80.3	83.9	62.9	86.0	83.5	23.4	84.2	22.3	84.1	6.6	180	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.1	89.8	76.7	89.2	76.4	88.1	76.0	79.8	84.8	79.1	84.4	78.8	84.2	14.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	149.7	82.4	77.4	142.9	81.5	77.2	141.8	81.3	43.5	85.2	42.8	84.8	42.2	84.9	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	21.1	19.2	DB	70.6	91.1	1.7	1.7	69.4	92.3	68.4	93.3	67.5	94.3	66.2	95.6	(4)
(5)			WB	65.1	81.9	1.7	2.5	63.8	83.7	62.8	85.1	61.9	86.5	60.6	88.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.6	82.2	82.0	81.5	81.2	81.4	80.8	80.5	80.8	81.5	82.1	82.5	82.9
	DBStd	1.77	2.03	2.19	2.08	1.84	1.57	1.56	1.50	1.29	1.25	1.21	1.17	1.37
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11538	998	895	976	935	974	924	946	954	944	995	976	1021
	CDD65	6063	533	475	511	485	509	474	481	489	494	530	526	556
	CDH74	62523	5765	5121	5316	4901	5253	4640	4556	4743	4927	5495	5660	6146
CDH80	18029	1757	1595	1592	1407	1546	1259	1141	1221	1321	1555	1693	1942	
Wind	WSAvg	10.2	9.6	8.6	7.3	7.1	8.1	9.1	10.4	12.1	13.3	13.1	12.5	11.4
Precipitation	PrecAvg	62.6	5.4	7.8	13.4	13.8	9.1	5.7	3.3	1.0	0.7	0.5	0.5	1.6
	PrecMax	111.9	26.3	19.8	32.7	26.3	26.4	22.4	11.7	7.5	5.8	2.2	1.9	8.3
	PrecMin	35.0	0.4	0.4	2.9	2.2	1.7	0.6	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	19.8	4.7	4.9	5.8	6.2	4.9	4.3	2.7	1.3	0.9	0.6	0.4	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	90.0	89.9	89.7	89.5	89.2	88.2	89.1	89.3	89.5	89.7	90.0
		MCWB	77.3	77.9	77.4	77.6	77.1	75.9	73.8	73.4	73.3	74.6	75.6	77.1
	2%	DB	88.7	89.4	88.3	88.1	88.1	87.8	87.7	87.8	87.9	88.1	88.2	89.2
		MCWB	76.9	77.5	77.5	77.4	77.1	75.5	73.9	73.3	73.6	74.4	75.4	76.7
	5%	DB	87.8	88.0	87.7	87.6	87.5	86.4	86.3	86.3	86.9	87.5	87.7	87.9
		MCWB	76.6	77.3	77.5	77.4	77.0	75.2	73.7	73.1	73.4	74.3	75.4	76.2
	10%	DB	86.4	86.6	86.3	86.1	86.2	85.9	85.7	85.8	86.0	86.2	86.4	87.1
		MCWB	76.3	77.1	77.3	77.2	76.6	75.1	73.7	73.0	73.3	73.9	74.8	75.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.4	81.4	80.9	80.0	79.8	78.8	77.8	77.8	77.3	77.7	78.2	79.1
		MCDB	85.7	85.8	84.8	85.0	85.2	84.5	82.5	82.8	83.2	84.6	85.4	86.3
	2%	WB	78.7	79.9	79.4	79.1	78.9	77.9	76.9	76.6	76.4	76.8	77.2	78.2
		MCDB	84.8	84.5	84.3	84.2	84.4	83.5	81.9	81.9	82.5	83.9	84.6	85.2
	5%	WB	78.1	79.1	78.9	78.7	78.3	77.3	76.3	75.9	75.6	76.1	76.6	77.4
		MCDB	84.1	84.1	84.0	83.7	83.7	82.8	81.5	81.4	81.9	83.1	84.0	84.2
	10%	WB	77.4	78.4	78.3	78.2	77.8	76.8	75.6	75.3	75.3	75.5	76.1	77.1
		MCDB	83.4	83.6	83.5	83.2	83.2	82.2	81.0	81.1	81.7	82.5	83.4	83.8
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.7	10.0	10.1	10.4	10.8	11.2	11.3	10.6	9.9	9.5	9.1
		MCDBR	10.1	10.4	10.9	11.4	11.3	11.5	11.9	11.9	11.2	10.4	9.9	9.7
		MCWBR	3.3	3.7	4.0	4.3	4.1	4.4	4.9	5.0	4.3	3.8	3.3	3.1
	5% WB	MCDBR	9.7	10.0	10.2	10.6	10.7	11.0	11.1	11.3	10.5	9.9	9.6	9.5
		MCWBR	4.0	4.5	4.7	4.6	4.3	4.4	4.7	4.9	4.2	4.0	3.4	3.4
Clear-Sky Solar Irradiance	taub		0.418	0.414	0.414	0.406	0.405	0.405	0.391	0.407	0.438	0.431	0.421	0.426
	taud		2.456	2.479	2.489	2.530	2.517	2.509	2.545	2.468	2.350	2.366	2.405	2.411
	Ebn at Noon		291	292	289	284	276	271	278	279	278	284	288	287
	Edh at Noon		37	37	36	34	33	32	31	35	41	41	39	39
All-Sky Solar Radiation	RadAvg		1766	1718	1664	1548	1629	1618	1730	1950	2070	2087	2042	1904
	RadStd		141	144	145	143	120	91	68	56	64	45	54	72

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page